

Reality Probe: Information-Limited Physical Verification of Captured Worlds

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Captured scenes are increasingly used as executable worlds whose physical parameters can be fitted, edited, and replayed. Lower observation error, however, does not by itself establish that a physical rewrite is more correct. We introduce Reality Probe, an evidence-governed rewrite-verification framework that separates proposal generation from a distinct verification channel and permits SUPPORT, VETO, or UNRESOLVED outcomes. A staged falsification program first shows why this separation matters: Dark-Field Counterfactual Spectroscopy (DCS), a mechanism-selective intervention contrast used inside the framework, succeeds on constructed controls but provides zero resolved coverage in several frozen synthetic regimes. Replacing the kinematic probe with known-force compliance raises median standardized evidence magnitude by 11.46 $\times$, yet still leaves all 36 fresh synthetic repair proposals unresolved at the frozen decision magnitude. We then test a better-matched real-video physical observable on a repository-locked IRIS pendulum final set of 10 videos. The finite-amplitude period probe makes 90/110 controlled decisions correctly versus 40/110 for the matched small-angle baseline, including 40/50 correct SUPPORT cases, 40/50 correct VETO cases, and 10/10 correct UNRESOLVED placebo cases. A second-domain IRIS free-fall replication subsequently fails untouched validation: all five videos pass low-level quality checks, but the median acceleration relative error is 111.7%, so that lane is retained as a negative result and its final split remains unopened. Reality Probe therefore does not provide universal physical certification; its effectiveness depends on whether the verification channel contains discriminating information for the requested rewrite, and the correct system action can be to abstain.

Additional Key Words and Phrases: captured worlds, inverse physics, physical verification, counterfactual simulation, material point method, uncertainty, experiment design, Gaussian splatting

1 INTRODUCTION

Recent systems increasingly attach physical state to visual scene representations so that captured objects can be simulated, edited, and interrogated under new interactions. Physics-aware neural fields and Gaussian representations have made this coupling practical across dynamic reconstruction, material inference, and interaction synthesis [Li et al. 2023; Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Xie et al. 2024; Zhang et al. 2024]. These systems raise a verification question that is distinct from fitting: after a candidate physical rewrite improves the observations used to assess it, what evidence establishes that the rewrite is actually more correct under interventions that were not part of the repair objective?

This distinction matters because inverse physical models are commonly underdetermined by the observations available to them. Different parameterizations, transfer schemes, boundary assumptions, numerical settings, or constitutive approximations can fit similar observed motion while making different predictions elsewhere. Work on structural identifiability, optimal experimental design, model discrepancy, and simulation credibility already provides mature tools for analyzing related ambiguity [Asadi and Laksari 2025; Laudy 2026; Oberkamp et al. 2007; Wang et al. 2026b]. Our focus is narrower and operational: the object being judged is a *requested physical rewrite*

of a persistent captured world, and the verifier must decide whether separately reserved evidence supports the rewrite, contradicts it, or is insufficient to decide.

We call a repair *deceptive* when it improves an ordinary held-out observation metric while worsening an untouched physical target. This is not a claim about intent or a particular optimizer. It is a failure mode of the evidence used to approve an edit. The selected case below shows one exact example from our fresh synthetic force-compliance experiment. The repair reduces ordinary held-out error by 14.93% while increasing the untouched-target error by 9.03%. The case is selected only for visualization; aggregate statistics and decision thresholds are independent of this selection.

We investigate this problem through *Reality Probe*, an evidence-governed verification pipeline for captured executable worlds. Candidate repairs are proposed using ordinary calibration and held-out observations. Verification then applies controlled physical interventions that were not used to choose the repair, compares the truth-like reference response with candidate responses, propagates measurement, repeat, and numerical uncertainty, and produces a support, veto, or unresolved decision. Reality Probe is the contribution-level framework: it separates proposal generation from evidence, carries explicit uncertainty, preserves support/veto/unresolved outcomes, quantifies the information frontier, and permits refusal. Dark-Field Counterfactual Spectroscopy (DCS) is one mechanism-selective instrument tested inside that framework. DCS constructs signed intervention contrasts that cancel lower-order common response so that residual disagreement becomes more visible. The operators used to build these contrasts are classical finite-difference and inclusion-exclusion constructions; our contribution is their use as a repair-verification instrument and, equally importantly, the empirical characterization of where that instrument fails.

The paper follows a falsification sequence rather than a positive-only benchmark. DCS first passes a constructed positive control. It then fails to produce resolved evidence on frozen truth-world validation, adaptive witness-space refinement, and a pair-specific repair-veto population. Matched raw, Fisher-style, and maximum-motion comparators likewise provide zero resolved coverage in the repair-veto population. We therefore quantify the *information frontier*: how far the observed standardized evidence lies from the frozen decision reference. Finally, we freeze a new orthogonal channel before execution, applying known external forces and measuring compliance. This change increases median standardized signal by 11.46 $\times$ on fresh synthetic truth worlds, yet still does not cross the fixed decision magnitude.

The resulting contribution is not a claim that DCS or force compliance is a solved prospective verifier. Instead, this work contributes:

1. **Rewrite-level verification contract.** We formalize deceptive physical repair and develop an evidence-governed transaction that separates proposal generation from a distinct verification channel, carries explicit uncertainty, and preserves



Fig. 1. Reality Probe at a glance, composed from frozen result panels. A candidate repair can improve ordinary held-out fit while worsening an untouched physical target. The repair is then asked a different physical question, its motion response is compared against the truth response, mechanism-level residuals are exposed, and the evidence is standardized. The selected case is post hoc for visualization; the 11.46× signal gain and 0/36 resolved coverage are aggregate fresh synthetic force-compliance results.

- SUPPORT, VETO, or UNRESOLVED outcomes rather than forcing a binary verdict.
- Information-boundary falsification ladder.** We show that lower-order-annihilating intervention contrasts can succeed on a constructed control yet fail to provide useful prospective discrimination in frozen regimes, and that changing to known-force compliance raises median standardized evidence by 11.46× without crossing the unchanged decision boundary.
- Repository-locked real-video confirmation.** On 10 fresh IRIS pendulum videos, a finite-amplitude period probe yields 90/110 correct controlled decisions versus 40/110 for the matched small-angle baseline, with all 10 placebo cases correctly left unresolved.
- Preserved second-domain failure.** An untouched IRIS free-fall validation set passes low-level tracking quality on 5/5

videos but fails the physical recovery gate with 111.7% median acceleration relative error. We retain this failure and leave the designated final split unopened instead of retuning the method on confirmatory data.

The central conclusion is therefore about evidence, not optimism: a captured world can look better and still be physically worse, and a verifier should remain unresolved when the available experiment does not contain enough discriminating information.

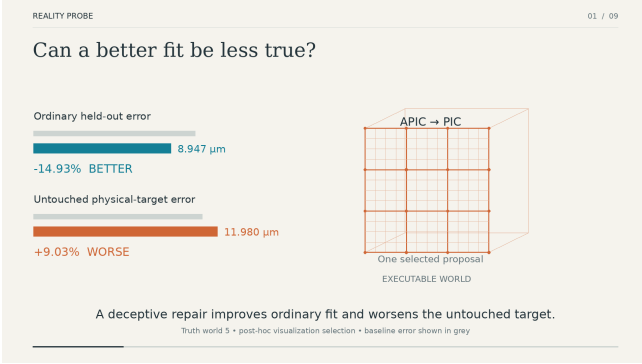


Fig. 2. Selected deceptive repair from the frozen fresh synthetic force-compliance population. The ordinary held-out error improves by 14.93% while the untouched physical-target error worsens by 9.03%. This post-hoc visualization case does not affect aggregate statistics or thresholds.

2 RELATED WORK

2.1 Physics-aware scene representations and inverse physics

PAC-NeRF couples continuum simulation to a neural radiance field for joint geometry and physical-parameter estimation from multi-view video [Li et al. 2023]. PhysGaussian integrates Material Point Method (MPM) dynamics directly with 3D Gaussian primitives [Xie et al. 2024]; subsequent work has extended physics-aware visual representations toward material initialization and refinement, interaction synthesis, and monocular or multi-material inverse physics [Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Zhang et al. 2024]. These methods establish that high-quality visual representations can support physical inference and simulation. Reality Probe does not claim novelty for Gaussian-plus-MPM simulation, parameter fitting, or differentiable inverse physics. We instead ask whether a proposed physical rewrite is supported by a verification channel once an optimizer has already produced an observational improvement.

Differentiable simulation systems such as ChainQueen and Diff-Taichi provide important infrastructure for gradient-based physical optimization [Hu et al. 2020, 2019]. In our implementation, finite differences, adjoint sensitivity, APIC/MPM replay, and material calibration are enabling components rather than contributions by themselves.

2.2 Identifiability and experimental design

Physical inference can be limited by the information content of the observed trajectories. Recent work provides structural identifiability conditions directly from video for classes of dynamical systems [Wang et al. 2026b], while measurement-grounded benchmarks such as IRIS evaluate parameter accuracy, identifiability, extrapolation, and governing-equation recovery on controlled real videos [Khanbayov et al. 2026]. In mechanics, optimal experiment design uses sensitivity, Jacobian conditioning, and Fisher-information concepts to choose informative loading conditions [Asadi and Laksari

2025]. These ideas motivate our matched baselines and the distinction between weak and informative interventions. Classical model-discrimination and optimal-experiment-design reasoning likewise asks which excitation best separates competing explanations; Reality Probe does not claim that principle as new. Our narrower contribution is to make evidence sufficiency an explicit transaction rule for a requested captured-world rewrite, to retain unresolved outcomes, and to measure what changes when the physical evidence channel itself is changed.

2.3 Simulation credibility and measured physical fidelity

Verification, validation, uncertainty quantification, and intended-use credibility have long been central to computational modeling [Oberkampff et al. 2007]. More recent digital-twin work explicitly considers validation of simulated counterfactuals [Laudy 2026]. These frameworks make broad credibility questions explicit, but they do not remove the application-specific need to determine whether a particular physical edit is supported by the available experiment.

GAUGE provides a recent measurement-grounded benchmark for physical fidelity across simulation engines and video world models, covering rigid, cable, textile, and volumetric-deformation tasks [Wang et al. 2026a]. We use a public GAUGE foam-shearing subset only for a retrospective channel-contradiction analysis. That analysis is intentionally not presented as fresh prospective confirmation.

A separate repository-frozen GAUGE validation lane uses previously untouched foam stretching and compression repeats 5–7. Across 12 measured validation worlds, the temporal annihilating witness makes 0/12 correct SUPPORT decisions and 0/12 correct VETO decisions while correctly leaving 12/12 placebo cases UNRESOLVED at the frozen decision magnitude. No final-test repeat 8–10 is opened. We retain this as prospective validation-stage evidence of insufficient directional decision coverage, not as a result to be repaired by threshold tuning.

2.4 Measured deformable tracking

For engineering validation of the captured-world pipeline, we also use sequence C2 from the DOT deformable tracking dataset, which provides measured 3D correspondences obtained using invisible markers [Li et al. 2025]. DOT supplies geometry and motion measurements, but not the stress-free state, loads, thickness, mass, or constitutive ground truth required to identify the true cloth material. We therefore treat the inferred parameters in that benchmark as model-conditioned effective quantities and keep this measured-source system check separate from the main verification claim.

3 PROBLEM FORMULATION

Let M denote an executable captured world and let R_0 and R_1 denote a baseline and a proposed repaired physical realization. We distinguish three data roles: fitting evidence used to calibrate or propose a rewrite, ordinary held-out evidence used to measure generalization within the same observation family, and an untouched target or separately reserved physical intervention used only after the repair is proposed.

Let $L_{obs}(R)$ be the ordinary held-out loss and $L_{target}(R)$ be an independent physical target. We define a deceptive repair by

$$L_{obs}(R_1) < L_{obs}(R_0) \quad \text{and} \quad L_{target}(R_1) > L_{target}(R_0).$$

This definition is deliberately agnostic to why the mismatch occurs. Model-form inadequacy, parameter compensation, transfer choice, boundary assumptions, and numerical artifacts can all create observationally plausible but physically different worlds.

A verification channel produces an error $e(R, Y)$ against reserved verification evidence Y and an uncertainty scale σ . We use the signed progress statistic

$$z = \frac{e(R_0, Y) - e(R_1, Y)}{\sigma}.$$

The frozen decision rule is

$$z \geq 2 \Rightarrow \text{support}, \quad z \leq -2 \Rightarrow \text{veto}, \quad |z| < 2 \Rightarrow \text{unresolved}.$$

The magnitude 2 is an operational decision reference fixed before the fresh evaluation; it is not claimed to be a universal physical constant. The distinction matters because the main experiments remain unresolved rather than being reclassified after inspection.

4 REALITY PROBE

4.1 Captured-world rewrite and verification pipeline

Reality Probe separates proposal generation from verification. A captured bundle provides appearance-linked state, particle state, observations, uncertainty sidecars, and provenance. The implementation calibrates a candidate physical model using the fitting split, evaluates ordinary held-out error, proposes a local or model-level rewrite, and only then evaluates separately reserved counterfactual evidence. If the verification channel produces support, the system may commit the rewrite; if it produces veto, the rewrite is rejected; if evidence is unresolved, the system refuses certification rather than forcing a binary answer.

The evidence layer deliberately has three outcomes, while the lower-level world transaction has two terminal states: committed or not committed. In the implementation, support can lead to a verified commit; both veto and unresolved map to a rejected/non-commit transaction, and any tentative mutation is rolled back. The three-way scientific distinction is retained in the evidence record even though the storage transaction is binary.

The verification transaction is deliberately sequential: proposal generation is completed before the separately reserved physical probe is evaluated, and the resulting evidence may support, veto, or leave the rewrite unresolved. This ordering prevents the confirmatory channel from silently becoming part of the proposal objective.

The research implementation is in C++20 and uses nonlinear MPM replay, APIC/PIC transfer variants, finite-difference and adjoint influence paths, deterministic experiment drivers, frozen JSON/CSV result ledgers, and SHA-256-indexed evidence packaging. The paper does not depend on native rendering for its quantitative conclusions; Metal and Vulkan render paths are used for system integration and presentation.

4.2 Dark-Field Counterfactual Spectroscopy

For a model M and physical intervention u , let $F_M(u)$ denote an observable response vector. DCS applies a signed intervention measure μ to form

$$\mathcal{D}_\mu[F_M] = \sum_i w_i F_M(u_i).$$

For a declared mechanism order k , the stencil weights are required to annihilate all lower-order multivariate monomials,

$$r_\alpha = \sum_i w_i u_i^\alpha \approx 0, \quad |\alpha| < k.$$

The implementation enumerates these monomials and validates the maximum residual against a tolerance rather than hard-coding only one stencil. A familiar order-2 one-dimensional example is

$$F(+a) + F(-a) - 2F(0). \quad (1)$$

which cancels constant and linear response. In the experiments, however, the order-2 comparator is synthesized on a 3×3 intervention lattice; Eq. 1 is explanatory rather than the literal selected stencil.

The system also implements finite-amplitude Boolean-lattice interactions through the Möbius contrast

$$\kappa(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} F(T).$$

We treat this as an interaction statistic, not as a newly derived mixed-derivative theorem.

4.3 Uncertainty and standardized separation

The evidence denominator separates measurement, repeat, and numerical uncertainty. Independent measurement and repeat variance propagate through a signed stencil with the squared ℓ_2 weight gain $\sum_i w_i^2$. Numerical error is treated more conservatively because discretization error can be correlated across intervention points. D3 additionally estimates numerical uncertainty directly in witness space by evaluating the same signed witness at nominal and refined solver resolution.

For candidate pairs, the verifier evaluates worst-case standardized separation over the retained response bundle. Active stencil synthesis first projects candidate disagreement into the nullspace of the lower-moment constraints and then searches deterministic projected directions. This is a heuristic for a non-convex maximin problem; the implementation does not provide a global-optimality certificate.

4.4 Known-force compliance as orthogonal information

After D4V, we freeze a separate experiment to test whether the low evidence magnitude is specific to the DCS construction or reflects a broader shortage of informative physical observations. The new channel leaves the fitting and repair-proposal path unchanged and applies a known force only after a candidate repair has been selected.

Six fresh synthetic truth worlds use APIC transfer with Young's modulus in $\{14125, 16625, 18125\}$ Pa and Poisson ratio in $\{0.22, 0.32\}$, with truth timestep 2.5×10^{-5} s. Candidate families are fixed before

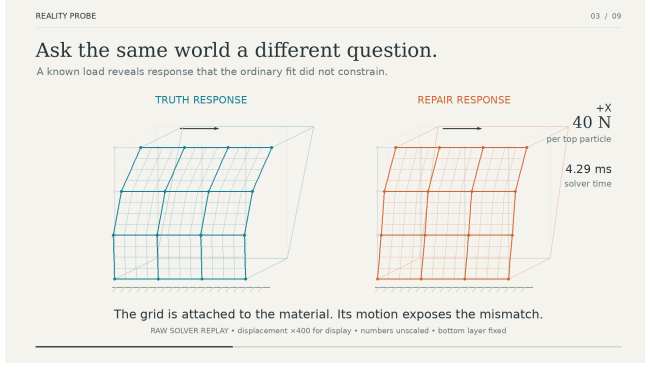


Fig. 3. Same controlled +X force probe applied to truth and repaired worlds. The displayed deformation is magnified 400x only for legibility; reported numbers remain unscaled.

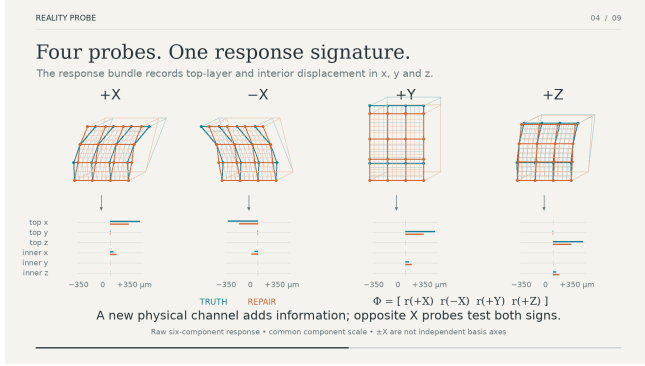


Fig. 4. Four-direction force-response fingerprint for the selected frozen case. The six-component response bundle records top-layer and interior displacement under +X, -X, +Y, and +Z probes; truth and repair are plotted on a shared component scale.

execution. The bottom particle layer is kinematically fixed, gravity is zero, and each top-layer particle receives 40 N in each of four experiments: +x, -x, +y, and +z. For every direction we record the mean top-layer and interior-layer displacement in three axes, yielding a six-component response per force experiment.

Numerical uncertainty is the RMS difference between nominal and half-timestep candidate response bundles. Measurement and repeat scales are each frozen at 2×10^{-5} m per component. The force progress statistic is

$$z_{force} = \frac{e(B, Y) - e(R, Y)}{\sqrt{\sigma_{meas}^2 + \sigma_{repeat}^2 + \sigma_{num,B}^2 + \sigma_{num,R}^2}}.$$

The same decision magnitude of 2 is used for the DCS comparator and force channel. Force amplitude, directions, truth worlds, candidate grid, and threshold are frozen before execution and are not returned after observing the results.

5 EXPERIMENTAL PROGRAM

Our experiments are organized to distinguish implementation sanity checks from frozen validation, discovery analyses, retrospective

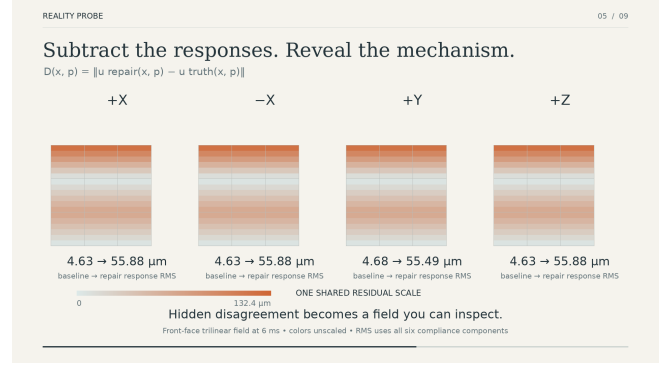


Fig. 5. Solver-driven residual field for the same frozen visualization case. All four force directions share one residual scale; the field is explanatory and is not itself the DCS decision statistic.

measured evidence, and fresh follow-on evidence. Table 1 summarizes the central outcomes.

Main experimental stages. “Crosses frozen boundary” means that the standardized evidence reaches the predeclared support/veto magnitude; zero crossings do not imply that an experiment produced no scientific information.

5.1 Graphics-facing benchmark and visual evidence coverage

A graphics submission needs recognizable benchmark evidence and inspectable visual consequences, not only an internal systems ledger. We therefore make the evidence surface explicit rather than asking the reader to infer it from the implementation. Table 2 separates public measured datasets, controlled synthetic evaluation, and visualization-only geometry. This classification does not change any experimental gate or elevate retrospective evidence into prospective confirmation.

The paper’s deceptive-repair, same-probe, force-fingerprint, residual-field, signal-gain, and information-limit panels are generated from committed solver trajectories or frozen result ledgers. They are intended to show what the tested physical differences look like, while the numerical conclusions continue to come from the JSON/CSV evidence records. In other words, the visual layer exposes the consequence of the experiments without becoming a second, informal source of truth.

5.2 D1: constructed positive control

D1 verifies the execution path under deliberately constructed deceptive cases. All 64 proposed repairs improve the ordinary criterion and are deceptive by construction. DCS rejects all 64. Median ordinary observation improvement is 60.88%, and the median witness-degradation ratio is 9.375. We use this stage only as an implementation control; it does not establish prospective generalization.

5.3 D2: active selection and frozen validation

An active-selection discovery comparison evaluates how well different witnesses rank target behavior. On D2b, ranking agreement is 95.83% for raw maximin, 87.50% for Fisher/Jacobian selection,

Table 1. Main experimental stages. “Resolved” refers to the frozen support/veto decision rule, not whether the scientific question was informative.

Stage	Population	Crosses frozen boundary	Main outcome
D1	64 repairs	64/64	Constructed positive control; median ordinary improvement 60.88%, median witness degradation ratio 9.375.
D2	16 worlds	0/16	Median standardized separation 0.041578.
D3	6 worlds	0/6	Median adaptive separation 0.0588583; lower numerical floor does not recover observability.
D4V	36 repairs	0/36	14 deceptive and 22 beneficial; matched verification channels also have zero resolved coverage.
GAUGE	10 trials	–	Retrospective measured channel contradiction; longitudinal endpoint preferred in 9/10 held-out repeats.
GAUGE validation	12 measured worlds / 36 cases	gate fail	Frozen stretching/compression validation: 0/12 correct SUPPORT, 0/12 correct VETO, and 12/12 correct UNRESOLVED placebo decisions; final repeats 8–10 remain unopened.
OFC	36 repairs	0/36	12 deceptive and 24 beneficial; known-force compliance increases evidence scale but not coverage.
IRIS pendulum	10 videos / 110 cases	90/110 correct	Locked final set; finite-amplitude probe gives 40/50 correct SUPPORT, 40/50 correct VETO, and 10/10 correct UNRESOLVED placebo decisions.
IRIS free-fall	5 validation videos	gate fail	Fresh second-domain validation has 5/5 quality-pass videos but 111.7% median acceleration relative error; final split remains unopened.

Table 2. Graphics-facing evidence coverage. Public measured benchmarks are kept distinct from controlled synthetic partitions and from visualization-only geometry.

Source	Evidence class	What it contributes	Claim boundary
DOT C2 [Li et al. 2025]	Public measured trajectories	End-to-end captured-world ingestion, fit/hold-out replay, local rewrite, and rollback on measured motion.	Stress-free state, loads, thickness, mass, and constitutive ground truth are not supplied; fitted material values remain model-conditioned.
GAUGE foam shearing [Wang et al. 2026a]	Public measured benchmark	Ten held-out repeats for a measured channel-contradiction analysis.	Retrospective/post-hoc evidence only; it is not fresh prospective confirmation of DCS.
D2/D3/D4V + OFC	Frozen controlled synthetic	Known-ground-truth falsification, pair-specific repair evaluation, and orthogonal force-compliance observability under a frozen decision rule.	Controlled evidence, not a measured-world benchmark.
Stanford Bunny visual suite	Public geometry visualization carrier	Solver-driven deformation, force direction, mechanism fingerprints, and residual fields for qualitative inspection.	Visualization only; the bunny is not benchmark geometry and contributes no quantitative result.

75.00% for the same-cost raw bundle, 66.67% for maximum motion,

62.50% for DCS order-2 maximin, and 45.83% for random selection. DCS therefore does not dominate simpler matched baselines.

Table 3. Ordinary-improving proposal populations. D4V and fresh synthetic OFC are independently generated partitions and their counts must not be mixed.

Population	Deceptive	Beneficial	Resolved
D4V	14	22	0/36
Fresh OFC	12	24	0/36

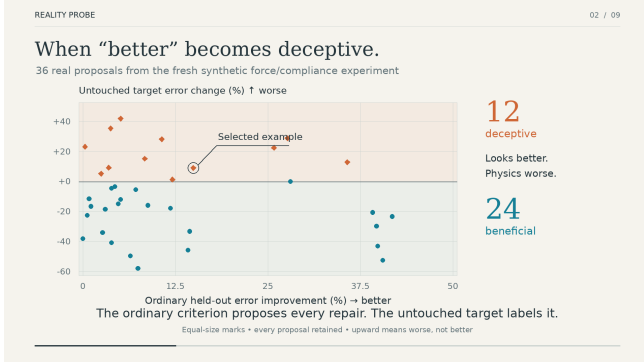


Fig. 6. Fresh orthogonal force-compliance proposal landscape. The figure shows where ordinary held-out improvement and the untouched physical target disagree across the fresh population.

The frozen D2 validation uses 16 truth worlds. No world reaches the decision magnitude. Median standardized separation is 0.041578, median numerical RMS is 6.469×10^{-6} m, and the maximum moment residual is 3.886×10^{-16} . The stencil algebra is therefore numerically well satisfied while the physical evidence remains weak.

5.4 D3: witness-space refinement and adaptive order

D3 tests whether direct witness-space numerical treatment or higher mechanism order recovers useful separation. Six fresh synthetic truth worlds are evaluated. Adaptive selection chooses order 2 in all six cases and never chooses order 3. Median adaptive separation is 0.0588583. Refinement lowers the median numerical RMS from 8.435×10^{-7} m for order 2 to 2.153×10^{-7} m for order 3, yet no truth world is resolved. Ranking agreement remains modest: Fisher 61.11%, raw bundle 58.33%, maximum motion 58.33%, pair-aware raw 55.56%, adaptive DCS 52.78%, and order-3 DCS 47.22%.

This stage separates two issues that are often conflated: numerical witness fidelity can improve without making the experiment more informative about the competing physical worlds.

5.5 D4V: pair-specific repair veto

D4V evaluates 36 repairs that already improve ordinary held-out error. Fourteen are deceptive according to the untouched target and 22 are beneficial. DCS, raw-bundle, pair-specific raw, Fisher, and maximum-motion verification all produce zero resolved coverage at the inherited $|z| \geq 2$ decision rule. The numeric table below reports the D4V and fresh follow-on proposal counts, followed by the fresh proposal landscape.

Table 4. Public DOT C2 measured-source engineering check. The material parameters are model-conditioned effective values, not material ground truth.

Quantity	Result
Measured correspondence points	225
Trajectory observations (fit / held-out)	675 (585 / 90)
Effective E, ν	7.5 kPa, 0.45
Dynamic RMS (fit / held-out)	4.391 / 4.417 mm
Adaptive candidate set	182/225 particles; 94.80% abs. gradient
Rewrite verification	nonlinear pass; oracle fail; rollback

A post-hoc information-frontier analysis quantifies the gap. In D4V the median DCS magnitude is 0.05024 against the frozen gap-magnitude 2 decision reference, corresponding to roughly 39.8x median signal amplification if uncertainty were otherwise unchanged. The matched channels are also far from resolved coverage. We treat this analysis as diagnostic only; it does not modify any frozen gate.

5.6 Measured-source engineering check

The end-to-end system is also exercised on measured DOT C2 trajectories [Li et al. 2025]. The importer keeps measured 3D correspondences separate from explicit model proxies for rest state, density, thickness, mass, Gaussian photometry, and uncertainty. Fit-only calibration and held-out replay produce a model-conditioned effective material setting rather than a claim of true cloth-property recovery. A proposed local material rewrite is rejected and rolled back because the complete independent derivative-verification contract is not met, despite satisfying the configured nonlinear linearization-error tolerance. This result validates the commit/refusal mechanism on measured-source data but is not used as evidence for the central deceptive-repair claim.

The same pinned DOT C2 source is also exercised by the native Vulkan measured-world showcase, which emits baseline and roll-back hero views, close-ups, a comparison plate, and an eight-view turntable. Because the candidate rewrite is rejected, the final displayed state intentionally returns to the baseline state. These images are qualitative implementation evidence for the measured captured-world path; they are not additional evidence that the fitted material parameters are physically unique or ground-truth accurate.

5.7 GAUGE: retrospective measured channel contradiction

We analyze ten held-out foam-shearing repeats from GAUGE [Wang et al. 2026a]. The committed paired-diagnostics CSV is channel-aggregated: each row reports win/loss/tie counts over these same ten repeats rather than representing a single trial. Ordinary face-plus-marker metrics prefer the finite-overlap candidate in all 10 repeats. Marker dark-field also prefers overlap in all 10. In contrast, the longitudinal mechanism channel prefers the endpoint model in 9 of 10 repeats. Median marker error is 0.002428 m for endpoint and 0.001709 m for overlap, whereas median longitudinal error is



Fig. 7. Held-out DOT C2 measured-versus-predicted geometry at the later validation checkpoint ($t = 0.167$ s), shown in the camera-frame x/y projection. Circles are measured correspondences, crosses are Vulkax predictions, and connecting segments are true-scale residuals rather than magnified arrows. Across the 90 held-out samples from both nonzero checkpoints, the 3-D RMS is 4.417 mm. This evaluates the model-conditioned captured-world replay; DOT C2 does not provide material ground truth.

Table 5. GAUGE retrospective channel comparison. Lower error is better. The sign test is post hoc.

Metric	Endpoint	Overlap
Median marker error (m)	0.002428	0.001709
Median longitudinal error	0.003131	0.005584
Longitudinal held-out wins	9/10	1/10

0.003131 for endpoint and 0.005584 for overlap. A post-hoc exact two-sided sign test on the longitudinal direction yields $p = 0.02148$.

The metric conflict was known before DCS was formulated. We therefore label this evidence retrospective throughout and do not use it as prospective confirmation.

5.8 Fresh orthogonal force-compliance study

The fresh synthetic force experiment yields 36 ordinary-heldout-improving proposals: 12 deceptive and 24 beneficial. Fresh DCS median $|z|$ is 0.04877 with maximum 0.17279. Force compliance

Table 6. Fresh synthetic orthogonal force-compliance results.

Quantity	DCS	Force
Median $ z $	0.04877	0.55863
Maximum $ z $	0.17279	1.31911
Resolved coverage	0/36	0/36
Median factor to $ z = 2$	41.01×	3.58×
Best-case factor to $ z = 2$	11.57×	1.52×

Table 7. Threshold-robustness audit from the frozen per-proposal ledger. The $\tau = 2$ row is the prospectively frozen primary rule; other rows are post-hoc sensitivity checks and do not redefine the decision policy.

Decision magnitude τ	DCS crossings	Force crossings
1.0	0/36	14/36
1.5	0/36	0/36
2.0	0/36	0/36
2.5	0/36	0/36
3.0	0/36	0/36

median $|z|$ is 0.55863 with maximum 1.31911. The ratio of aggregate medians is 11.455. Despite the substantial increase, DCS resolves 0/36 and force compliance resolves 0/36. The median additional amplification needed for the force channel to reach magnitude 2 is 3.58x; the best case still requires 1.52x.

Decision-threshold robustness. The primary decision magnitude remains the prospectively frozen $\tau = 2$. To check whether the central negative conclusion is merely an artifact of that particular number, we also inspect nearby thresholds without changing the primary rule. The frozen per-proposal OFC visualization ledger retains the complete signed DCS and force progress scores, so the diagnostic sweep can be reported exactly. Fresh DCS crosses neither support nor veto even at $\tau = 1$. At $\tau = 1$, force compliance resolves 14/36 proposals: 6 support and 8 veto. Five of the 12 deceptive repairs are vetoed (41.7% recall), while 3 of 24 beneficial repairs are falsely vetoed (12.5%). The result therefore remains below the repository-frozen advancement target even under this looser post-hoc threshold. At $\tau = 1.5$ and above, force coverage returns to 0/36. This robustness check leaves the primary conclusion unchanged: under the frozen $\tau = 2$ rule the fresh population remains uncertified, and the observed transition between $\tau = 1$ and $\tau = 1.5$ makes the threshold dependence explicit rather than hiding it.

The ratio of the aggregate median evidence magnitudes is 11.455x. The full post-hoc sweep is generated from the frozen per-proposal ledger by `research/analysis/threshold_sensitivity_ofc.py` and stored as `research/paper_data/THRESHOLD_SENSITIVITY_OFc_2026-09-21.csv`.

The protocol includes an advancement gate requiring at least 30% resolved coverage, at least 50% veto recall over deceptive repairs, no more than 25% false veto rate over beneficial repairs, better coverage than DCS, higher median magnitude than DCS, and finite numerical refinement for every proposal. The force experiment satisfies the population-size, false-veto, median-signal, and numerical checks but

fails the coverage and recall requirements. The result is therefore retained as a negative gate with a strong standardized-evidence increase, not relabeled as a successful verifier.

5.9 Locked IRIS pendulum confirmation

The synthetic ladder establishes that verification quality is controlled by the information channel, but it does not by itself provide measured-world confirmation. We therefore froze a real-video pendulum protocol before opening the designated final media. The final set contains 10 IRIS pendulum_90 videos, all of which pass the quality gate. The primary probe uses the finite-amplitude pendulum period relation; the matched diagnostic baseline uses the small-angle approximation. The proposal/candidate schedule, decision magnitude, tracker, and final-test media are fixed by repository state before the final run.

Across the 110 controlled cases on those 10 physical videos, the finite-amplitude probe makes 90 decisions correctly (81.82%), compared with 40/110 (36.36%) for the small-angle baseline. The primary probe is correct on 40/50 SUPPORT cases and 40/50 VETO cases, and it leaves all 10 placebo cases UNRESOLVED as intended. In paired comparison, the primary probe is correct when the baseline is not on 50 cases; the reverse occurs on zero cases, with 60 ties. These 110 rows are repeated controlled cases nested inside 10 physical videos and are not 110 independent experiments. The locked physical-recovery diagnostic has a median period-inferred length relative error of 16.81%.

This result does not show that finite-amplitude pendulum physics is new, nor that Reality Probe is universally reliable. It shows that, when the requested rewrite is interrogated through a physical observable whose information matches the distinction being tested, the same SUPPORT/VETO/UNRESOLVED transaction can produce useful decisions on fresh measured video.

5.10 Post-final hardening of the pendulum result

After the locked final test was complete, we ran secondary analyses that are not allowed to change its decisions. Treating the physical video rather than the nested case row as the experimental unit, the finite-amplitude probe has higher per-video accuracy than the small-angle baseline on all 10 videos (10–0, no ties; two-sided exact sign-test $p = 0.001953125$). Two stronger post-final comparators also fail to explain the locked result: a direct finite-amplitude period-residual rule reaches 45.45% strict three-way accuracy, while a damped nonlinear ODE residual reaches 9.09%, compared with the locked probe’s 81.82%. In paired case accounting, the locked probe is uniquely correct on 40 cases relative to the direct residual and on 80 relative to the damped ODE; neither post-final comparator is uniquely correct on any case.

We also remove the manually scheduled candidate factors in a separate ground-truth-hidden transaction. Baseline and candidate estimates are produced from early video windows, ordinary improvement is checked on a later holdout, and the blind proposal artifact is SHA-256 closed before ground truth is joined. All 10 generated proposals improve the ordinary holdout; 8 are physically worse and 2 are physically better, and the reserved final-window verifier classifies all 10 correctly. Canonical labels remain unchanged

over the recorded rope-length uncertainty intervals through $\pm 2\sigma$. A post-final proposal/probe overlap sweep also remains correct for every tested overlap, from disjoint windows through full overlap; this diagnoses temporal-window dependence but does not make same-video evidence an independent sensor modality.

Finally, a full Metal corruption sweep processes 180 condition-video runs with zero failed videos and reproduces the clean locked decision vector on all 10 videos. The 81.82% clean accuracy is unchanged under the tested additive noise (up to 30 intensity levels), Gaussian blur (kernel 15), frame duplication/drop (up to 50%), central occlusion (up to 30%), and crop (up to 10%). Temporal downsampling is the clear boundary: strict accuracy falls to 63.64% at 30 fps and 45.45% at 15 fps, although placebo abstention remains 100%. These tests are post-final stress analyses, not new confirmation, and the temporal-resolution failure is retained as a limitation.

5.11 Second-domain IRIS free-fall failure

We also attempted a second, different equation-family replication on IRIS dropping-ball video. The V6.6 selector was frozen after passing its development gate on drop_50: 4/5 quality-pass videos, zero implementation errors, and 8.05% median acceleration relative error. We then opened the untouched drop_100/06 . . 10 validation split without changing the selector.

The prospective result failed decisively. All 5/5 validation videos pass the low-level quality checks, but the median acceleration relative error is 111.7%; the truth-control accuracy and direction-sign rate are both 0 under the frozen decision logic. Post-validation diagnosis shows that the selector often bound the full known drop height to a shorter partial ballistic interval. Retrospective rescue experiments with alternative event ranking, higher image resolution, longer association, track stitching, global/chunk scale, stationary endpoint scale, and bidirectional hold-to-hold reconstruction did not recover a robust target-free estimator across the spent development and validation evidence.

We therefore close this lane as a negative result. The failed drop_100 split is permanently retrospective, no rescue result is treated as confirmatory evidence, and the designated drop_150 final split remains unopened. The failure reinforces rather than weakens the paper’s central boundary: a verification transaction is only as useful as the physical information actually available to it.

6 RESULTS AND ANALYSIS

6.1 Deceptive repair is a distinct failure mode

The D4V and fresh synthetic OFC populations establish that lower ordinary held-out error does not uniquely determine whether the untouched physical target improves. In D4V, 14 of 36 ordinary-improving repairs are deceptive. In the fresh synthetic OFC population, 12 of 36 are deceptive. These counts are not intended as estimates of a universal deception rate; they establish that the failure mode occurs under controlled candidate-generation pipelines and must therefore be considered by any verification rule that approves physical edits from observation fit alone.

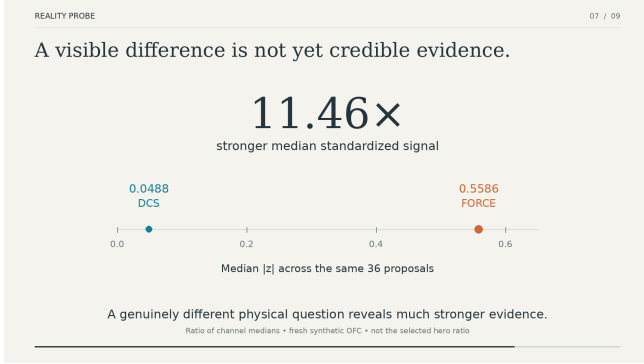


Fig. 8. Fresh known-force compliance raises the median standardized signal by 11.46x relative to fresh DCS without changing the frozen decision rule.

6.2 The stronger DCS hypothesis is not supported in the tested regime

The constructed D1 control demonstrates that the implementation can produce a strong dark-field rejection when the experiment is engineered to contain that distinction. The subsequent stages do not preserve that strength. D2 and D3 remain far below the decision magnitude, DCS does not dominate matched ranking baselines, and D4V provides zero resolved coverage. These negative results are central rather than incidental: they rule out the stronger interpretation that mechanism-selective cancellation alone is sufficient for repair certification in the tested regime.

The D3 numerical refinement result is particularly informative. Reducing the witness-space numerical floor by several-fold does not materially increase standardized separation. The limiting factor is therefore not simply solver discretization error in the original witness evaluation.

6.3 Orthogonal physical information changes the evidence scale

The information-limit result below summarizes the frozen evidence scale. The known-force compliance channel moves the median from the same order as the frozen D2/D3/DCS evidence to 0.5586. This is an order-of-magnitude change in the observed standardized evidence scale under a genuinely different physical interrogation, not a threshold adjustment or a new score applied to the same observations. We interpret this as strong evidence that observability depends on the experiment, while avoiding the stronger claim that information content is the unique limiting cause.

This result supports the interpretation that verification quality depends strongly on which physical question is asked. Kinematic observations that are adequate for proposing a plausible repair need not be adequate for discriminating that repair from alternatives. A force-controlled experiment adds information that is not available to the fitting objective.

6.4 Stronger evidence is not sufficient evidence

The strongest result in the fresh synthetic force population is $|z| = 1.31911$, still below the decision magnitude. The selected visual example has signed force progress $z = -1.31811$: evidence moves

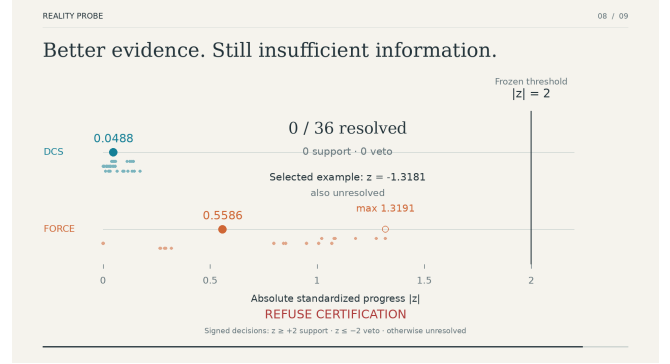


Fig. 9. Information-limit summary from the frozen fresh synthetic orthogonal-information experiment. Known-force compliance increases standardized evidence substantially, but the strongest case remains below the frozen decision magnitude.

substantially in the veto direction but remains unresolved. The system therefore refuses certification. We view this behavior as preferable to lowering the threshold after observing the partition or interpreting an intermediate score as a definitive physical verdict.

The information-frontier view also avoids a misleading binary narrative. The force channel is not equivalent to DCS simply because both return 0% resolved coverage. Their evidence magnitudes are materially different, and the remaining gap to the decision rule is correspondingly different. The experiment therefore supplies a quantitative direction for future sensor and intervention design even though it does not support deployment-level verification.

7 DISCUSSION

7.1 What is established

The experiments support four linked conclusions. First, observationally improving physical repairs can be deceptive with respect to an untouched target. Second, mechanism-sensitive verification signals can remain too weak to support a reliable repair decision even when their numerical implementation is stable. Third, changing the physical information channel can materially change observability: known-force compliance raises the synthetic evidence scale, and the locked finite-amplitude pendulum observable produces useful SUPPORT/VETO/UNRESOLVED decisions on fresh real video. Fourth, that success does not automatically transfer across domains: the frozen free-fall replication fails untouched validation despite passing its development gate.

These conclusions are narrower than a universal physical-correctness claim, but they are operational. A persistent captured-world system can preserve the distinction between fitting, verification, and unresolved evidence, can commit or veto a rewrite when a matched channel provides sufficient evidence, and can refuse or retire a verification lane when it does not.

7.2 Relation to identifiability and experiment design

The force-compliance result is consistent with the broader principle that parameter and model discrimination depend on excitation and measurement design [Asadi and Laksari 2025; Wang et al. 2026b].

Our experiments do not introduce that principle. Their contribution is to expose its consequence inside a physical-rewrite workflow: verification can fail not because a candidate looks implausible, but because the available experiment does not separate plausible candidates strongly enough. This distinction changes the system response from “pick the best score” to “request more informative evidence or remain unresolved.”

7.3 Relation to simulation verification and digital twins

Simulation credibility frameworks already separate code verification, solution verification, model validation, and uncertainty [Oberkampf et al. 2007]. Digital-twin validation work similarly emphasizes which counterfactual claims are supported by observable validation data [Laudy 2026]. Reality Probe should therefore be read as an application-specific mechanism rather than a replacement for those frameworks. Its scope is a physically editable captured world where a candidate rewrite is proposed and a separately defined verification intervention is used to decide whether the rewrite should be committed.

7.4 Limitations

The main limitations are substantive.

- **Fresh prospective force evidence is synthetic.** The orthogonal force-compliance experiment uses controlled MPM truth worlds rather than a measured force-sensor setup. A real prospective force experiment is required for a stronger measured-world claim.
- **GAUGE is retrospective.** The measured channel contradiction motivates the problem and demonstrates that different observables can disagree, but it is not an untouched prospective test of DCS or force compliance.
- **Measured positive evidence currently comes from one physical regime.** The locked IRIS pendulum result contains 10 fresh videos from one `pendulum_90` regime. Controlled case rows are nested within those videos and must not be treated as independent physical experiments.
- **Pendulum verification is temporally resolution-sensitive.** In the full post-final corruption sweep, tested noise, blur, frame duplication/drop, occlusion, and crop preserve the clean 81.82% strict accuracy, but temporal downsampling reduces it to 63.64% at 30 fps and 45.45% at 15 fps. Placebo abstention remains 100%. We therefore do not claim robustness to arbitrary acquisition frame rates or temporal sampling.
- **The second real-video domain failed.** The frozen IRIS free-fall replication fails untouched validation with 111.7% median acceleration relative error despite 5/5 low-level quality passes. That failure is retained, its validation split is permanently retrospective, and the designated final split remains unopened.
- **The physical regime is limited.** The principal experiments use a fixed deformable MPM setup and selected APIC/PIC candidate families. Generalization across constitutive families, contact regimes, and solvers is not established.
- **The decision magnitude is operational.** The value 2 is frozen before evaluation and prevents post-hoc threshold

tuning, but it is not a universal law for all physical verification tasks. A post-hoc robustness audit uses the frozen per-proposal ledger: force compliance has 14/36 crossings at $\tau = 1$ (6 support and 8 veto), while both fresh channels have zero crossings for every $\tau \geq 1.5$.

- **DCS synthesis is heuristic.** Lower-moment annihilation is exact up to tolerance, but the projected maximin search is not globally optimal.
- **The response-jet interpretation is operational.** The implementation estimates order of contact from validated witnesses; it does not reconstruct a complete symbolic Taylor tensor.
- **The selected hero case is post hoc.** It is chosen deterministically for explanation and does not influence thresholds, aggregate statistics, or the standardized-evidence result.
- **Spatial residuals are explanatory.** Solver-driven residual fields localize response disagreement but are not themselves the DCS decision statistic.
- **Measured DOT parameters are model-conditioned.** The DOT benchmark lacks stress-free state, loads, thickness, mass, and constitutive ground truth for C2; fitted material values are therefore not presented as true cloth properties.

7.5 Future work

The most direct next step is not another retuning of the frozen DCS, force-compliance, or failed free-fall partitions. It is a separately frozen measured experiment whose physical observable is selected before final data access and whose information content is plausibly matched to the rewrite being tested. Independent force sensing, calibrated excitation, modal or frequency-domain response, richer constitutive families, and intervention-specific sensor design are plausible directions. Any such experiment should keep proposal generation blind to the confirmatory channel, preserve a predeclared SUPPORT/VETO/UNRESOLVED rule, and retain failed domains without recycling them into confirmatory evidence.

8 REPRODUCIBILITY AND IMPLEMENTATION NOTES

The repository records the scientific freeze dated 2026-09-21 at commit `a9da8c0aa868`. The one-command runner builds and tests the code, executes the staged experiments, regenerates deterministic figures and tables, validates frozen numerical values, captures compiler/system/GPU provenance, and assembles a SHA-256-indexed evidence archive. Scientific conclusions are read from machine-readable JSON ledgers rather than from rounded figure labels.

The core DCS implementation is separated from experiment drivers and includes arbitrary-order lower-moment validation, signed response application, finite-amplitude counterfactual cumulants, standardized discrepancies, uncertainty propagation, active stencil synthesis, witness-space numerical refinement, and spatial localization. The orthogonal force protocol and fresh synthetic truth worlds are versioned separately from the earlier D2/D3/D4V partitions. This separation is used to prevent a negative gate from being silently replaced by a retuned result on the same data.

Generative-AI-assisted research workflow. OpenAI ChatGPT was used during research for code drafting and debugging, experiment-orchestration and analysis scripts, visualization/manuscript tooling, and manuscript revision. AI-generated material was not treated as empirical evidence. Experimental partitions, thresholds, executed outputs, numerical claims were checked against committed research artifacts, and cited sources were checked against publication records before inclusion in the manuscript.

Paper-facing visualizations are deterministic. Solver-driven surfaces may use an explicit display magnification to make micrometer-scale motion visible; reported numerical values are unscaled. The Stanford Bunny, where used in presentation assets, is a visualization carrier rather than benchmark geometry and requires source attribution. The companion mathematical explainer uses the actual solver body and does not require the bunny.

9 CONCLUSION

A physically editable captured world should not equate better observation fit with better physics. Across controlled repair populations, we find ordinary-improving rewrites that worsen an untouched physical target. Reality Probe provides the evidence-governed transaction around such rewrites; DCS is one tested mechanism-selective instrument inside it rather than the contribution by itself. Frozen D2, D3, and D4V experiments show that a mathematically valid probe can still be information-poor, while known-force compliance shows that changing the physical channel can raise the evidence scale by 11.46 $\times$ without yet reaching the decision boundary. On a repository-locked set of 10 fresh IRIS pendulum videos, a better-matched finite-amplitude observable yields 90/110 correct controlled decisions versus 40/110 for the matched small-angle baseline and preserves all 10 placebo abstentions. The subsequent free-fall replication fails untouched validation and is retained as a negative result rather than rescued by post-validation tuning. The practical conclusion is therefore conditional: verification can support or veto a physical rewrite when the available experiment carries the right information, and a trustworthy system should remain unresolved—or retire the verification lane—when it does not.

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